

1. a) What is *linked list*? What are the advantages and disadvantages of linked list over the linear array. 2 CO
OR

b) What is *circular header list*? Write the advantages of circular header list over ordinary linked list. 3 CO
Let LIST be a *sorted linked list*. Write an algorithm to search an element from the LIST. What is the time complexity of this algorithm?

OR

Let LIST be a *linked list* of integers in memory. Write a procedure to find the maximum MAX of the values in LIST. What is the time complexity of this procedure?

c) Given is the following sorted linked list where the array INFO contains a list of integer numbers and LINK, START and AVAIL are the pointer fields. 2 CO

	1	2	3	4	5	6	7	8	9	10	11	12
INFO:			41	129		9	123	78	194	231	62	145
LINK:	2	5	11	12	0	3	4	7	10	0	8	9
START : 6 AVAIL : 1												
Determine the changes in the list if number XX is <i>added</i> to the list and then 145 is <i>deleted</i> from the list. [Here XX is the last two digits of your ID]												

d) What is *two way linked list*? Form a *two way linked list* from the one way list in Q 2(c). 3 CO

Group B sp24

2. a) What is *linked list*? What are the advantages and disadvantages of linked list over the linear array? 2 CO1 C1

OR

a) What is *circular header list*? Write the advantages of circular header list over ordinary linked list. 2 CO1 C1

b) You are given a linked list containing an odd number of elements. You don't know how many numbers there are in that list. Write an algorithm or C/C++ code to find the *middle element* of this list. 4 CO3 C2

OR

b) You are given a two-way linked list containing an odd number of elements. You don't know how many numbers there are in that list. Write an algorithm or C/C++ code to find the middle element of this list by iterating only once over the list. 4 CO3 C2

c) Consider the *alphabetized* list of patients in Fig-2c. Determine the changes in the data structure if
i) Ford is added to the list and then ii) Lane is deleted from the list. 2 CO1 C2

					START 5						AVAIL 10		
	1	2	3	4	5	6	7	8	9	10	11	12	
NAME	Kirk		Dean	Max	Adam		Lane	Green	Samu		Fields	Nelso	
Next	7	6	11	12	3	0	4	1	0	2	8	9	

Fig-2c

d) A DEQUE is a data structure that allows insertion and deletion of elements from both ends that allows the following four operations on it: 2 CO1 C2

1. PUSH_FRONT(X): This function adds an item X in front of its elements.
2. PUSH_BACK(X): This function adds an item X at the end of its elements.
3. POP_FRONT(): This function deletes the first element (if there is any).
4. POP_BACK(): This function deletes the last element (if there is any).

Suppose you are building a DEQUE where the underlying data structure would be a one way linked list. What would be the *time complexity* of these four operations? Explain.

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2. Suppose LIST is a linked list in memory consisting of numerical values. Write a procedure for each of the following: 5 CO1 C2
- Finding the maximum even number of the values in LIST
 - Finding the sum of even numbers that are less than or equal to N of the LIST (Note: N is the last two digits of your ID. i.e. if your ID is C151216, then N=16).

OR

Write a procedure which adds a given ITEM of information at the end of a list. 5 CO1 C2

- b) The following figure is a list of five hospital patients and their room numbers. 5 CO1 C2
- Fill the values for NSTART and NLINK so that they form an alphabetical listing of the names.
 - Fill the values for RSTART and RLINK so that they form an ordering of the room numbers.

NSTART

RSTART

(Note: Your last name is X of your ID. i.e. if your ID is

	NAME	ROOM	NLINK	RLINK
1	X	650		
2	Smit	422		
3	Adams	704		
4	Jones	N		
5	Burns	632		

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and N is the last three digits C151216, then N=216).

2. a) What is *linked list*? What are the advantages and disadvantages of linked list over the linear array. 2 CO1 C1

OR

What is *circular header list*? Write the advantages of circular header list over ordinary linked list.

- b) Let LIST be a *sorted* linked list. Write an algorithm to *search* an element from the LIST. 3 CO3 C2

OR

Let LIST be a *linked list* of integers in memory. Write a procedure to find the *maximum* MAX of the values in LIST.

- c) Given is the following sorted linked list where the array INFO contains a list of integer numbers and LINK, START and AVAIL are the pointer fields- 2 CO1 C2

	1	2	3	4	5	6	7	8	9	10		
11	12											
INFO:			41	129		9	123	78	194	231	62	145
LINK:	2	5	11	12	0	3	4	7	10	0	8	9

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START: 6 AVAIL: 1

Determine the changes in the list if number XX is added to the list and then 194 is deleted from the list.

[Here XX is the last two digits of your ID]

- d) What is *two way linked list*? Form a two way list from the one way list in Q 2(c). 3 CO1 C3